

Error-Mitigated Circuits for Variational Quantum Algorithms

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- **The Hadamard Test**
- **Error Mitigation Techniques**
 - **Zero-noise Extrapolation (ZNE)**
 - **Echo Verification (EV)**
 - **Virtual Distillation (VD)**
 - **Learning-Based Methods (ML-QEM)**
- **Results**
- **Mitigating Feedforward Circuits (PROM)**
- **Future Work**

A generalized phase estimation scheme that extracts information without the need of state tomography.

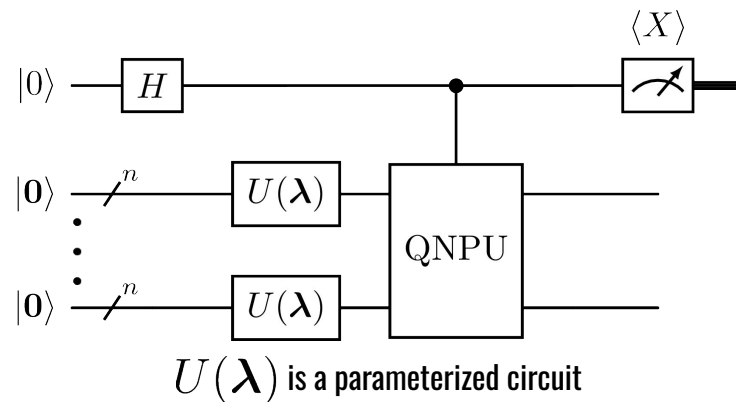
$$\begin{aligned}\mathcal{C}(\boldsymbol{\lambda}) &= \| |\Psi_{\text{target}}\rangle - |\Psi_{\text{var}}(\boldsymbol{\lambda})\rangle \|^2 \\ &= 2 \left[1 - \Re \left(\langle 0 | U_{\text{target}}^\dagger U(\boldsymbol{\lambda}) | 0 \rangle \right) \right]\end{aligned}$$

Error Mitigation (EM)

- Algorithmic level
- Mitigate the effect of errors via post-processing

Error Suppression

- Lower level
- Requires heavy device characterization
- eg dynamical decoupling, Pauli twirling



Error mitigation reduces the bias while increasing the variance compared to the noisy estimator.

Zero-Noise Extrapolation (ZNE)

Steps

- Amplify noisy in a well-defined way
- Extrapolate in post-processing
- Zero-noisy estimate via extrapolation

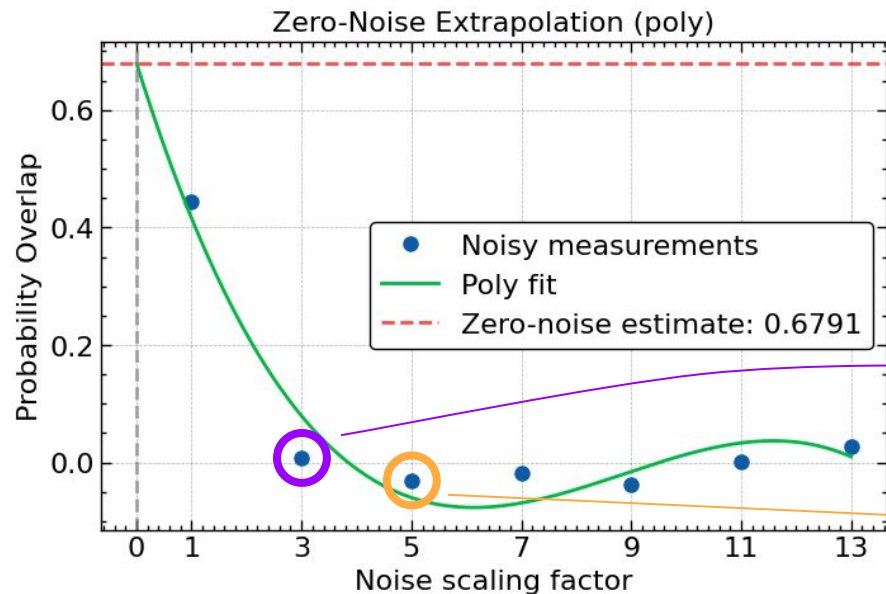
$$\langle \hat{O} \rangle_{\text{em}} = f(0; \vec{\theta}_{\text{opt}})$$

Pros

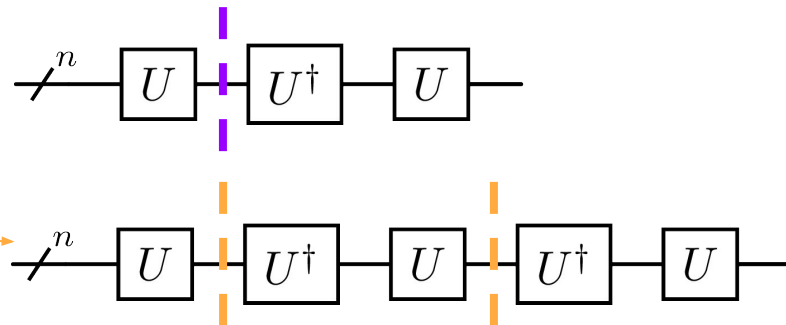
- Circuit and observable agnostic

Cons

- Sampling overhead for noise amplification



Circuit folding as a practical approach to noisy amplification

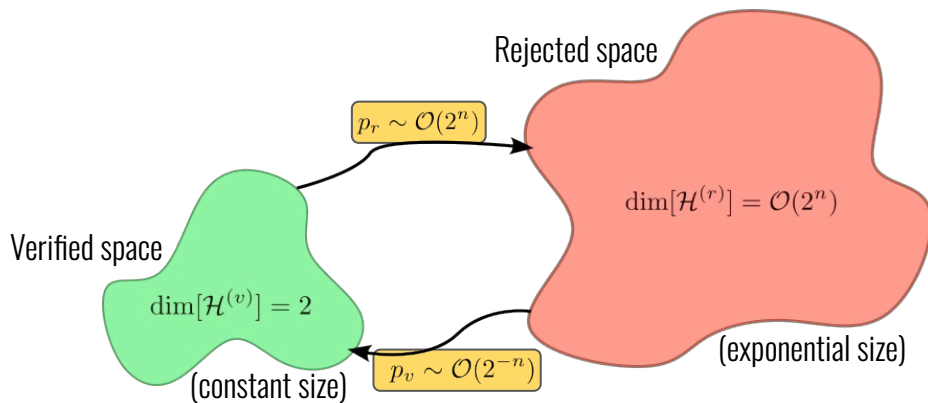


Echo Verification (EV)

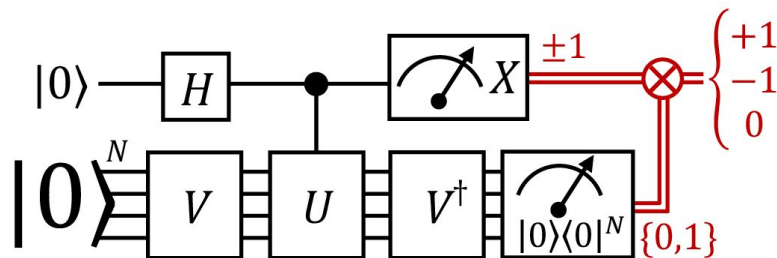
Steps

- Prepare the register with V
- Apply U
- Measure register qubits
- Post-select based on register measurements

On completion of the algorithm the $n+1$ qubits can be in two subspaces:



After the noise channel the probability of ending-up in $\mathcal{H}^{(r)}$ grows exponentially with the number of qubits.



M : # shots

N : # qubits

d : depth

ϵ : precision

p : single error rate

$$M \sim \epsilon^{-2} (1 - p)^{-N \cdot d}$$

Pros

- Great performance
- Noise agnostic
- Minimal circuit overhead

Cons

- Exponential shots with circuit size

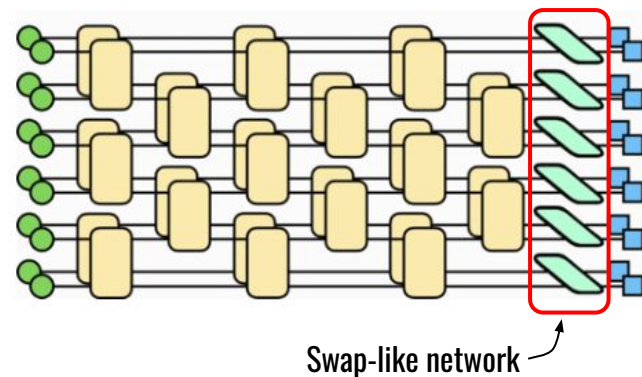
Approximate the closest pure state by entangling M copies

$$\frac{\rho^{\otimes M}}{\text{tr}(\rho^{\otimes M})} = \frac{\sum_i p_i^M |i\rangle \langle i|}{\sum_i p_i^M}$$

Converges exponentially with M

Error mitigation via *purification* requires additional implementations of the circuit in parallel.

This conflicts with the exponential encoding capability which is the main speedup in CFD use cases.



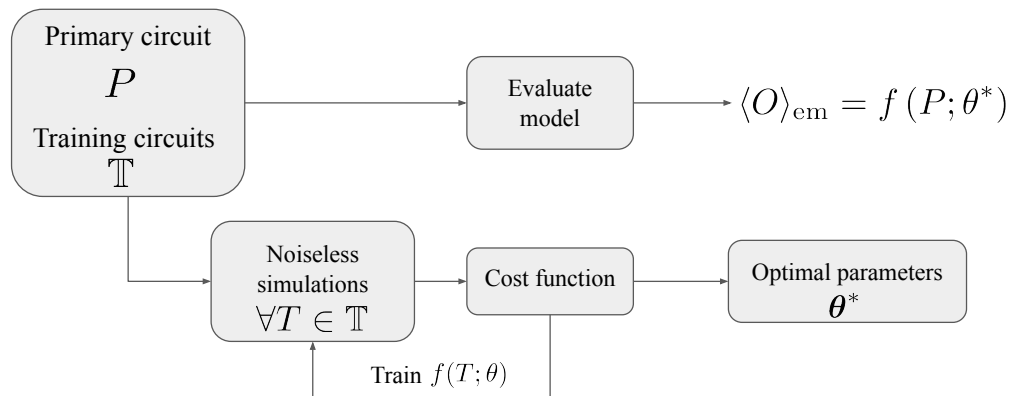
Pros

- No shot overhead

Cons

- Large ancilla overhead

Train a classical model on a noiseless dataset and evaluate the model on the noisy results



The desired circuit P might not be classically simulatable

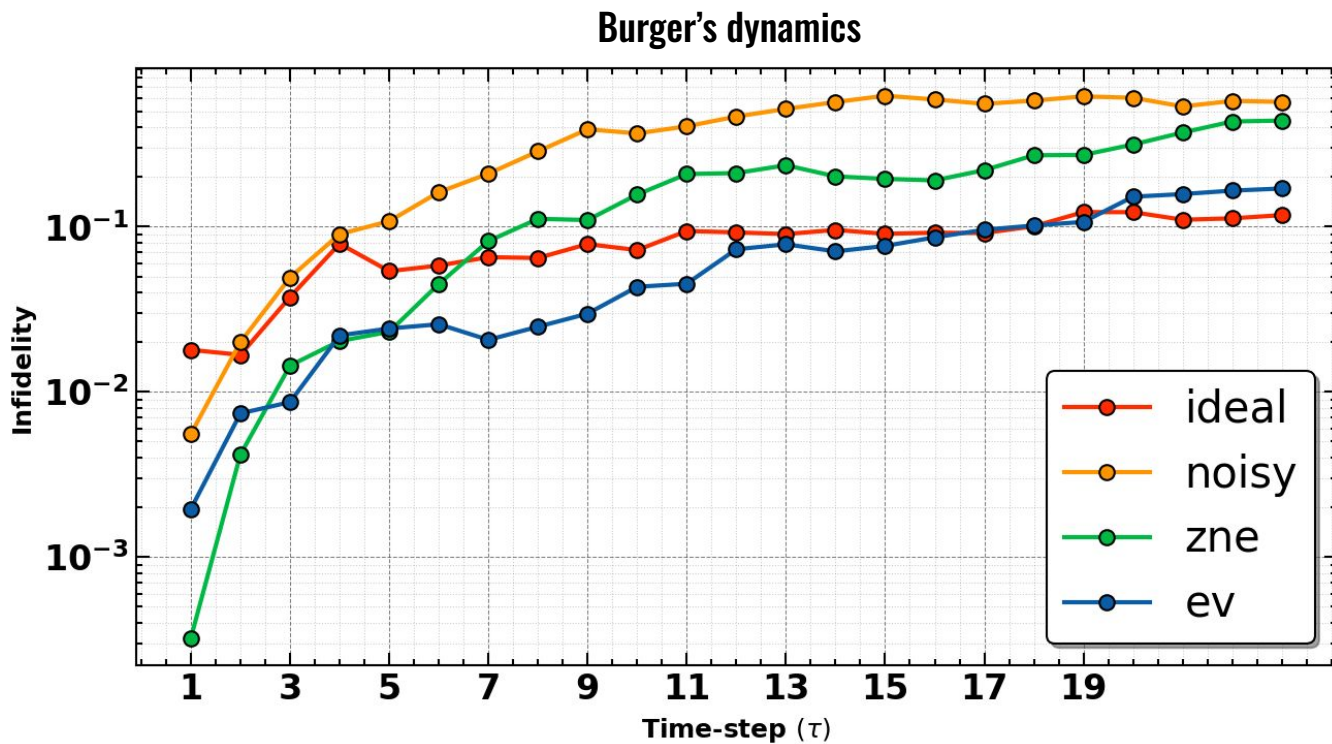
The model is trained over an ensemble of circuit primitives $\mathbb{T}$ that resemble P in with respect to the effects of noise

Pros

- Compatible with all methods

Cons

- Training overhead
- High-performance computing integration



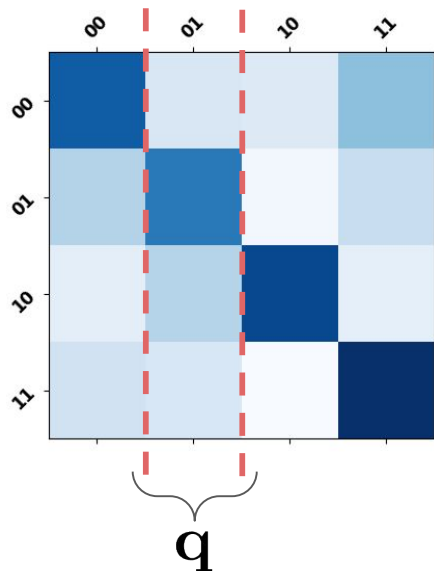
[Caveat: work in progress]

Probabilistic Readout Error-Mitigation (PROM)

Based on the groups prior work: circuits with mid-circuit measurements and feed-forwarding demonstrate noise resilience under assumptions

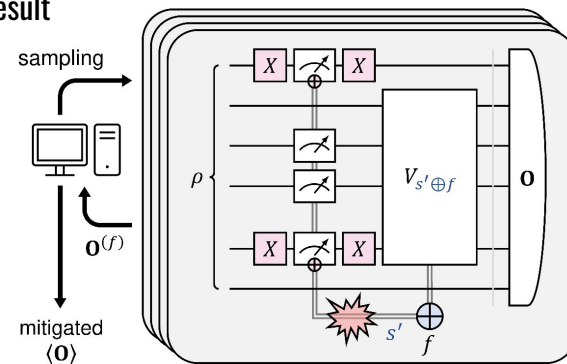
Probabilistically samples from an engineered ensemble of trajectories and computes the average result in post-processing.

Symmetrize errors by leveraging bit-flip averaging



Engineer some distribution α based on $\mathbf{q}$ and sample bitstring $f \in S$ with it.

Average over many runs to get the estimate.

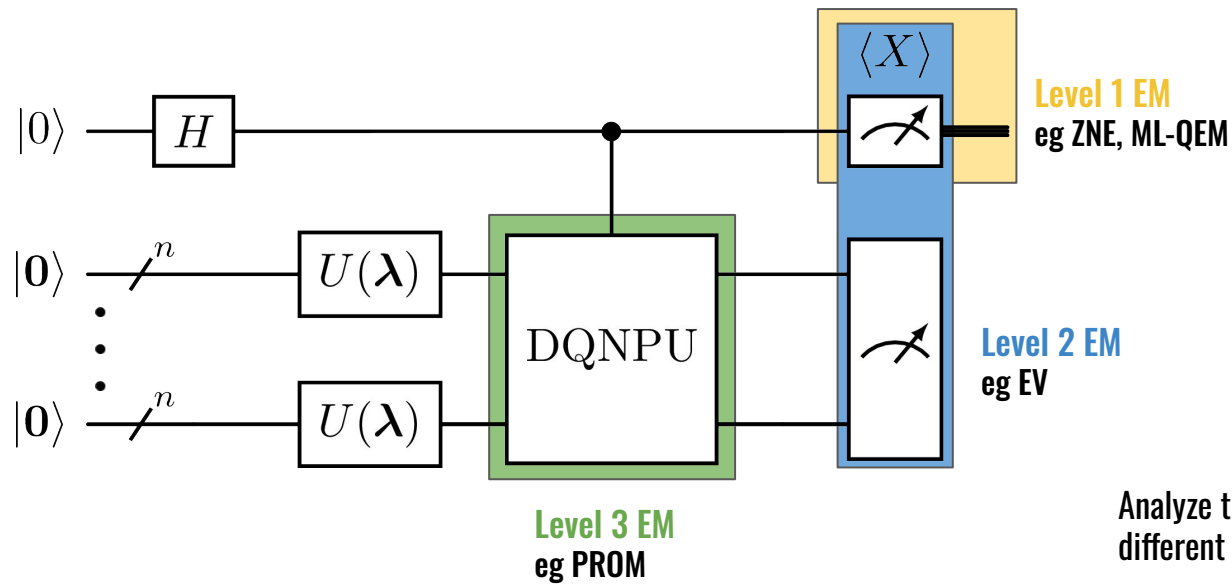


Pros

- Unbiased estimator
- Compatible with all other methods

Cons

- Sampling overhead depending on noise correlation



Analyze the performance and resources for different combination of multi-level QEM

Study the error mitigation capabilities of circuits with mid-circuit measurements and ancilla qubits